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Inventor(s)

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Skin care composition

Abstract

Low-pH, aqueous skin care composition, comprising: a) about 0.1% to about 10% of a hydroxycinnamic acid (HCA); b) about 0.1% to about 10% of a polar emollient; c) from about 1.5% to less than 20% of a co-solvent with a Hansen solubility parameter distance of less than 15 from the—hydroxycinnamic acid (HCA); and d) water, wherein the pH of the composition is about 5.0 or less, and wherein the composition meets at least one of the following conditions (i)-(iii): (i) the composition further contains from about 0.1% to about 10% of a vitamin B.sub.3 compound; (ii) the weight ratio between HCA and the polar emollient is from about 1:1 to about 1:10; and (iii) the composition further contains from about 0.1% to about 5% of a sensory improving ingredient selected from the group consisting of: sugar alcohols, sugar-based emulsifiers, modified sugar derivatives, and mixtures thereof.

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Background/Summary

FIELD

(1) The present disclosure generally relates to a low-pH, aqueous skin care composition, comprising: a) about 0.1% to about 10% of a hydroxycinnamic acid (HCA); b) about 0.1% to about 10% of a polar emollient; c) from about 1.5% to less than 20% of a co-solvent with a Hansen solubility parameter distance of less than 15 from the—hydroxycinnamic acid (HCA); and d) water, wherein the pH of the composition is about 5.0 or less, and wherein the composition meets at least one of the following conditions (i)-(iii): (i) the composition further contains from about

0.1% to about 10% of a vitamin B.sub.3 compound; (ii) the weight ratio between HCA and the polar emollient is from about 1:1 to about 1:10; and (iii) the composition further contains from about 0.1% to about 5% of a sensory improving ingredient selected from the group consisting of: sugar alcohols, sugar-based emulsifiers, modified sugar derivatives, and mixtures thereof.

BACKGROUND

(2) Skin is the first line of defense against environmental insults that would otherwise damage sensitive underlying tissue and organs. Additionally, skin plays a key role in a person's physical appearance. Not surprisingly, most people would like to have healthy, younger looking skin. Unfortunately for some people, the tell-tale signs of aging such as thinning skin, wrinkles, and age spots are an undesirable reminder of the disappearance of youth. The desire for healthy, younger looking skin has led to the development of numerous skin care products marketed to treat the various skin conditions associated with aging and unhealthy skin. These skin care products typically include one or more active ingredients for treating a skin condition of interest.

(3) Antioxidants are commonly used in skin care products as active ingredients, for example, as disclosed in WO 2008/73684. And hydroxycinnamic acids (HCAs) such as ferulic acid are particularly well-known antioxidants, for example, as described in WO 2018/081790. However, the use of hydroxycinnamic acids in skin care composition can be problematic. For example, HCAs are relatively insoluble in water, which can lead to the undesirable formation of HCA crystals in the product. Further, when HCA is used at neutral pH (e.g., pH 5.0-8.0), which is common for skin care compositions, the HCA may oxidize or degrade, causing undesirable color changes, odors and/or reduced efficacy of the product. In some instances, formulating the composition at a lower pH may help stabilize the HCA, but it also lowers HCA solubility.

(4) In order to increase the solubility of the HCA, some formulators add the HCA to the oil phase of the composition (i.e., in the case of an emulsion) or encapsulate the HCA. But these approaches can also be problematic. For example, adding a hydroxycinnamic acid to the oil phase can undesirably affect the sensory profile of the composition due to the introduction of oils and additional emulsifiers to the composition. And encapsulation can reduce the amount of hydroxycinnamic acid in the composition due to encapsulation loading limits. Thus, there remains a need to provide stabilized hydroxycinnamic acid in an aqueous composition at the desired level without undesirably affecting the sensory properties of the composition.

(5) L'oreal WO2021125070A1 discloses a composition comprising: (a) at least one cinnamic acid derivative such as ferulic acid; (b) at least one lipophilic antioxidant; (c) at least one oil such as isopropyl lauroyl sarcosinate; and (d) water, wherein the pH of the composition is 4.5 or less. L'oreal WO2021125070A1 also discloses in Examples, compositions comprising 0.5% ferulic acid, 6% of isopropyl lauroyl sarcosinate, 2% of dipropylene glycol, 10% of butylene glycol. There is still a need for such compositions to provide improved skin appearance and/or improved sensory.

SUMMARY

(6) Disclosed herein are low-pH, aqueous skin care compositions, comprising: a) about 0.1% to about 10% of a hydroxycinnamic acid (HCA); b) about 0.1% to about 10% of a polar emollient; c) from about 1.5% to less than 20% of a co-solvent with a Hansen solubility parameter distance of less than 15 from the—hydroxycinnamic acid (HCA); and d) water, wherein the pH of the composition is about 5.0 or less, and wherein the composition meets at least one of the following conditions (i)-(iii): (i) the composition further contains from about 0.1% to about 10% of a vitamin B.sub.3 compound; (ii) the weight ratio between HCA and the polar emollient is from about 1:1 to about 1:10; and (iii) the composition further contains from about 0.1% to about 5% of a sensory improving ingredient selected from the group consisting of: sugar alcohols, sugar-based emulsifiers, modified sugar derivatives, and mixtures thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A and 1b show the HPLC reference spectrum for HCA and 4-VP, respectively.
- (2) FIG. 2 illustrates HCA solubility in water across a range of pH. Chemicalize was used for prediction of HCA solubility (CAS #501-98-4, 7400-08-0), November 2021, <https://chemicalize.com/>, developed by ChemAxon.

DETAILED DESCRIPTION

(3) Hydroxycinnamic acids (HCAs) such as coumaric acid tend to exhibit undesirable solubility and/or stability characteristics in conventional aqueous skin care products. The insolubility of HCA can be even worse in a low-pH composition. It was recently discovered that low-pH compositions can improve the efficacy of certain skin care actives such as niacinamide (see, U.S. Pat. No. 10,874,600). Thus, it would be particularly advantageous to improve HCA solubility and/or stability in such compositions.

(4) Surprisingly, it has now been discovered that the stability and/or solubility characteristics of certain HCA compounds can be improved in a low-pH aqueous composition by formulating the composition with a suitable combination of polar emollient and glycol co-solvent. In particular, certain combinations of HCA, polar emollient and a co-solvent with a Hansen solubility parameter distance of less than 15 from the HCA can act synergistically to combat the undesirable stability and/or solubility effects of the HCA in an aqueous skin care composition.

(5) Reference within the specification to “embodiment(s)” or the like means that a particular material, feature, structure and/or characteristic described in connection with the embodiment is included in at least one embodiment, optionally a number of embodiments, but it does not mean that all embodiments incorporate the material, feature, structure, and/or characteristic described. Furthermore, materials, features, structures and/or characteristics may be combined in any suitable manner across different embodiments, and materials, features, structures and/or characteristics may be omitted or substituted from what is described. Thus, embodiments and aspects described herein may comprise or be combinable with elements or components of other embodiments and/or aspects despite not being expressly exemplified in combination, unless otherwise stated or an incompatibility is stated.

(6) In all embodiments, all percentages are by weight of the cosmetic composition, unless specifically stated otherwise. All ratios are weight ratios, unless specifically stated otherwise. All ranges are inclusive and combinable. The number of significant digits conveys neither a limitation on the indicated amounts nor on the accuracy of the measurements. All numerical amounts are understood to be modified by the word “about” unless otherwise specifically indicated. Unless otherwise indicated, all measurements are understood to be made at approximately 25° C. and at ambient conditions, where “ambient conditions” means conditions under about 1 atmosphere of pressure and at about 50% relative humidity. All numeric ranges are inclusive of narrower ranges; delineated upper and lower range limits are interchangeable to create further ranges not explicitly delineated.

(7) The compositions of the present invention can comprise, consist essentially of, or consist of, the essential components as well as optional ingredients described herein. As used herein, “consisting essentially of” means that the composition or component may include additional ingredients, but only if the additional ingredients do not materially alter the basic and novel characteristics of the claimed compositions or methods. As used in the description and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

Definitions

(8) “About” modifies a particular value by referring to a range equal to plus or minus twenty percent (+/-20%) or less (e.g., less than 15%, 10%, or even less than 5%) of the stated value.

(9) “Apply” or “application,” as used in reference to a composition, means to apply or spread the

compositions of the present invention onto a human skin surface such as the epidermis.

(10) “Cosmetic agent” means any substance, as well any component thereof, intended to be rubbed, poured, sprinkled, sprayed, introduced into, or otherwise applied to a mammalian body or any part thereof to provide a cosmetic effect. Cosmetic agents may include substances that are Generally Recognized as Safe (GRAS) by the US Food and Drug Administration, food additives, and materials used in non-cosmetic consumer products including over-the-counter medications.

(11) “Effective amount” means an amount of a compound or composition sufficient to significantly induce a positive benefit to keratinous tissue over the course of a treatment period. The positive benefit may be a health, appearance, and/or feel benefit, including, independently or in combination, the benefits disclosed herein. In a specific example, an effective amount of a vitamin B.sub.3 compound is an amount sufficient to improve the health and/or appearance of psoriatic skin during a treatment period. In some instances, an effective amount may be demonstrated using ex vivo and/or in vitro methods.

(12) “Hydroxycinnamic acid” (HCA) refers to a class of aromatic acids or phenylpropanoids having a C6-C3 skeleton that are hydroxy derivatives of cinnamic acid. Some non-limiting examples of HCA are caffeic acid, chicoric acid, cinnamic acid, chlorogenic acid, diferulic acids, coumaric acids (p-, o-, and m-isomers), ferulic acid, sinapinic acid, sinapic acid, and α -cyano-4-hydroxycinnamic acid.

(13) “Improve the appearance of” means providing a measurable, desirable change or benefit in skin appearance, which may be quantified, for example, by a decrease in redness, inflammation, and/or plaque scales.

(14) “Low-pH” means a pH of less than 5.0 (e.g., 1.5 to 4.9, 2.0 to 4.5, 2.5 to 4.0, or 3.5 to 4.0). A suitable method of determining the pH of a composition is described in more detail below.

(15) “Neutral pH” means a pH of 5.0 to 8.0.

(16) “Safe and effective amount” means an effective amount of an ingredient that is low enough to avoid serious side effects (within the scope of sound medical judgment).

(17) “Skin care” means regulating and/or improving a skin condition. Some nonlimiting examples include improving skin appearance and/or feel by providing a smoother, more even appearance and/or feel; increasing the thickness of one or more layers of the skin; improving the elasticity or resiliency of the skin; improving the firmness of the skin; and reducing the oily, shiny, and/or dull appearance of skin, improving the hydration status or moisturization of the skin, improving the appearance of fine lines and/or wrinkles, improving skin exfoliation or desquamation, plumping the skin, improving skin barrier properties, improve skin tone, reducing the appearance of redness or skin blotches, and/or improving the brightness, radiancy, or translucency of skin.

(18) “Skin care active” means a compound or combination of compounds that, when applied to skin, provide an acute and/or chronic benefit to skin or a type of cell commonly found therein. Skin care actives may regulate and/or improve skin or its associated cells (e.g., improve skin elasticity, hydration, skin barrier function, and/or cell metabolism).

(19) “Skin care composition” means a composition that includes a skin care active and regulates and/or improves skin condition.

(20) “Treatment period,” as used herein, means the length of time and/or frequency that a material or composition is applied to a target skin surface.

(21) Composition

(22) The skin care compositions described herein are low-pH compositions intended for topical application to human skin to improve the appearance, health, and/or function of skin. The present compositions may be used for non-therapeutic (i.e., cosmetic) treatment of a variety of skin conditions. For example, the low-pH composition may be particularly suitable for improving the appearance of skin, especially fine lines, wrinkles, hyperpigmented spots, uneven skin tone, sallow looking skin, and/or skin stability. Improved skin stability means to provide reduced day-to-day fluctuation of skin condition.

(23) The low-pH skin care compositions herein include a safe and effective amount of hydroxycinnamic acid (e.g., p-coumaric acid a.k.a. 4-HCA or p-HCA), a polar emollient and a suitable co-solvent with a Hansen solubility parameter distance of less than 15 from the HCA. The compositions herein may also include a vitamin B.sub.3 compound such as niacinamide and/or one or more other optional skin care actives. The composition may optionally include a silicone emulsifier, a polymer thickener that can tolerate low-pH environments, a low molecular weight silicone fluid, an acid-salt pH-buffering system (e.g., a lactic acid/sodium lactate buffering system), and/or other ingredients commonly found in topical skin care compositions. It is believed, without being limited by theory, that the combinations of ingredients disclosed herein provides a stable and efficacious skin care composition that has good feel properties and is gentle on skin.

(24) The compositions herein are formulated to provide improved HCA solubility in a low-pH environment. In an aqueous, low-pH skin care composition, HCA has a tendency to precipitate out of the composition and form crystals (HCA crystals), which can undesirably affect the look, feel, and/or efficacy of the composition. Conventional low-pH compositions often contain HCA crystals. However, the low-pH compositions herein contain a polar emollient and a co-solvent with a Hansen solubility parameter distance of less than 15 from the HCA. This combination of ingredients is specifically tailored to solubilize the HCA in the low-pH aqueous composition and inhibit HCA crystallization/precipitation. Thus, the compositions herein are free of HCA crystals. A suitable method for measuring and/or characterizing the presence of HCA crystals in a composition is described in more detail below.

(25) The low-pH compositions herein are also formulated to provide improved HCA stability. HCAs are relatively good antioxidant materials, and thus tend to be oxidized and/or degrade over time, which can result in a skin care composition that exhibits an undesirable color change (e.g., yellowing), an undesirable odor, and/or reduced efficacy. Formulating at a lower pH can improve HCA stability by reducing the rate at which it is oxidized, but it may be desirable, in some aspects, to include an antioxidant in the low-pH composition to help further reduce oxidation and/or degradation of the HCA.

(26) The low-pH skin care compositions herein can be made by mixing the ingredients with a dermatologically acceptable carrier using conventional methods known to those skilled in the art. The compositions may be provided in various product forms such as solutions, suspensions, lotions, creams, gels, toners, sticks, sprays, aerosols, ointments, cleansing liquid washes and solid bars, pastes, foams, mousses, shaving creams, wipes, strips, patches, electric-powered patches, hydrogels, film-forming products, facial and skin masks (with and without insoluble sheet), and the like. The composition form may follow from the particular dermatologically acceptable carrier chosen.

(27) In some instances, the low-pH skin care composition herein may be in the form of an essence. An essence is a form of topical skin care composition in a relatively concentrated formula that typically has a lower viscosity than a conventional cream- or lotion-type skin care composition. An essence may be provided in the form of a low viscosity fluid that is marketed to specifically target a particular skin condition and/or be used in the first step of a skin care regimen. The skin care essence products herein can have a dynamic viscosity of 1 centipoise (cP) to 15,000 cP at 25° C. (e.g., 50 cP to 10,000 cP or 100 cP to 7,500 cP, 200 cp to 5,000 cp, or 300 cp to 2,500 cp). A method of determining the viscosity of the low-pH compositions is described in more detail in the Methods section below.

(28) Hydroxycinnamic Acid

(29) The low-pH skin care compositions herein include a safe and effective amount of an HCA. The HCA may be present in the composition at 0.1% to 10% (e.g., 0.5% to 5% or 1% to 4%). Hydroxycinnamic acids are generally recognized as antioxidant phenolic compounds, which can be found in plants, mainly as a component of cell walls. See, H. K. Kuzaki et al., J. Agric. Food Chem., 50, 2161-68 (2002). In some aspects, it may be desirable to select coumaric acid for use in

the low-pH composition, especially p-coumaric acid (a.k.a. 4-HCA). 4-HCA has the formula:

(30) ##STR00001##

(31) A particularly suitable example of an HCA material suitable for use herein is LIPOBRITE available from Vantage Personal Care.

(32) Low-pH Solubilizer

(33) The compositions herein include one or more solubilizers and/or co-solvents to help solubilize the HCA. HCA compounds generally exhibit especially poor solubility in low-pH aqueous composition, such as the low-pH compositions described herein. For example, p-coumaric acid has a solubility of approximately 345 mg/mL in water at pH 7.0 and 20° C. and a solubility of 4 mg/mL in water at pH 3.0 and 20° C. Without being limited by theory, it is believed that the decreased solubility of an HCA at lower pH is due to reduced ability of the HCA to undergo the acid dissociation that forms the base species observed at higher pH levels. At pH 7, p-coumaric acid exists at approximately 99.6% in its base form; whereas at pH 3, the base form is only present at approximately 13.4%. As a result of its relatively poor solubility, HCAs can form crystals in an aqueous, low-pH skin care composition. The HCA crystals may impart an undesirable feel to the composition during use (e.g., a rough or grainy feel). HCA crystal formation may also lead to a decrease in the efficacy of the HCA and/or other ingredients in the composition, and/or create an undesirable consumer perception of the skin care composition or manufacturer (e.g., poor quality).

(34) In order to avoid the formation of HCA crystals in the present low-pH compositions, the HCA should have a solubility greater than the intrinsic solubility limit of the composition. For example, a 1.0% HCA formula at pH of 3.8 with water at approximately 75% has a concentration of 13.3 mg/ml at 20° C., whereas the intrinsic solubility is 6.8 mg/mL. In this example the HCA would likely crystallize. Hydroxycinnamic acids have been shown to complex and co-crystallize with certain materials, and so the intrinsic solubility limit alone may not fully predict crystallization stability. See, Bevill, et al., Polymorphic Cocrystals of Nutraceutical Compound p-Coumaric Acid with Nicotinamide: Characterization, Relative Solid-State Stability, and Conversion to Alternate Stoichiometries, Cryst. Growth Des. 2014, 14, 3, 1438-1448; 2014.

(35) It has now been found that certain combinations of polar emollients (e.g., isopropyl lauroyl sarcosinate) and co-solvents with a Hansen solubility parameter distance of less than 15 from the HCA (e.g., certain glycols) can greatly improve the solubility of HCA in a low-pH aqueous composition. Emollients are substances that tend to soften and moisturize skin by forming an oily layer on top of the skin that traps water in the skin. Some common examples of emollients include petrolatum, lanolin, mineral oil and dimethicone. Polar emollients are useful herein for solubilizing HCA, but polar emollients can also destabilize an oil-in-water emulsion, which is a common product form for skin care composition. Thus, it can be important to tailor the type and amount of polar emollient in the composition to help solubilize the HCA in the composition, but avoid emulsion instability. Not all polar emollients will work with all HCAs.

(36) In some aspects, the low-pH compositions herein may include from about 0.1% to about 10% of a polar emollient, and preferably such that the weight ratio between HCA and the polar emollient is from about 1:1 to about 1:10.

(37) In some aspects, the low-pH compositions herein may include 0.5% to 7% of a polar emollient (e.g., 1% to 5% ELDEW SL-205 brand isopropyl lauroyl sarcosinate available from Ajinomoto OmniChem), which has been found to solubilize p-coumaric acid, especially when combined with a suitable glycol. Some non-limiting examples of polar emollients that may be suitable for use herein are amino acid ester derivatives (e.g., lauroyl arginine, arginine lauroyl glutamate, phytosteryl/octyldodecyl lauroyl glutamate, and dihexyldecyl lauroyl glutamate) and hydrocarbon esters (e.g., cyclohexylglycerin and isopropyl myristate).

(38) The compositions herein also include a co-solvent with a Hansen solubility parameter distance of less than 15 from the HCA that acts in combination with the polar emollient to help solubilize the HCA. The Hansen solubility parameter distance is described in *Hansen solubility parameters: a*

user's handbook, CRC press, 2007 and can be calculated using the following equation:
$$\sqrt{\{(\delta_{D2}-\delta_{D1})^2+(\delta_{P2}-\delta_{P1})^2+(\delta_{H2}-\delta_{H1})^2\}}$$
 Where: δ_D =Hansen dispersion, δ_P =Hansen polarity, and δ_H =Hansen hydrogen bonding.

(39) Co-solvents suitable for use herein include short chain dihydric alcohols (e.g., glycols). However, when a glycol is selected as the co-solvent, it can be important to limit the amount of glycol in the low-pH composition to less than 25% (e.g., less than 20%, less than 17%, less than 15%, or even less than 10%), but generally greater than 1%, to avoid undesirable feel characteristics (e.g., sticky feeling or greasy feeling). Some non-limiting examples of glycols that may be suitable for use herein are propylene glycol, dipropylene glycol, butylene glycol, pentylene glycol, hexylene glycol, ethoxydiglycol, and C2-C6 polyethene glycols (e.g., PEG-3, PEG-4, PEG-4 methyl ether), and combinations thereof.

(40) In some aspects, the polar emollient and co-solvent may be present in the low-pH composition at a weight ratio of polar emollient to co-solvent of 1:3 to 3:1 (e.g., 2:3, 1:1, or 2:1). In some aspects, the HCA may be premixed with the co-solvent and/or polar emollient and, optionally, one or more other ingredients and then added to the composition.

(41) Sensory Improving Ingredients

(42) When the compositions contain higher amount of polar emollient, co-solvent, and mixtures thereof, such compositions may have an unpleasant consumer sensory experience for various attributes, including stickiness, oiliness, greasiness, coated residue feeling, shininess, and lack of absorption into skin. Thus, in such compositions containing higher amounts of polar emollients, co-solvents, and mixtures thereof, it is preferred to add sensory improvement ingredients. Such sensory improving ingredients useful herein included: sugar alcohols such as trehalose and xylitol; sugar-based emulsifiers such as sucrose dilaurate, and sorbitan stearate; and modified sugar derivative such as, sodium gluconate. Surprisingly, it was found these materials which were known to provide negative sensory perception such as stickiness, actually counter-acted the polar emollient and co-solvent effect in this case. Among them, sugar alcohols are more preferred in view of providing additional skin conditioning benefit together with sensory improving benefit.

(43) It is preferred that the sensory improving ingredient is included in the composition at a level of from about 0.1% to about 5%. It is also preferred that the weight ratio between sensory improving ingredient and co-solvent is from about 2:1 to about 1:20, more preferably from about 1.5:1 to about 1:10, still more preferably from about 1.5:1 to about 1:5. Vitamin B₃ compound

(44) The present composition may include a safe and effective amount of a vitamin B₃ compound for regulating a variety of skin condition, for example, as described in U.S. Pat. No. 5,939,082. The compositions herein may contain 0.1% to 10%, by weight, of the vitamin B₃ compound, based on the weight or volume of the composition (e.g., 0.5% to 5% or 1% to 4%).

(45) As used herein, "vitamin B₃ compound" means a compound having the formula:

(46) ##STR00002##

Where: R is CONH₂ (i.e., niacinamide), COOH (i.e., nicotinic acid) or CH₂OH (i.e., nicotinyl alcohol); derivatives thereof; and salts of any of the foregoing. Exemplary derivatives of vitamin B₃ compounds include nicotinic acid esters, including non-vasodilating esters of nicotinic acid (e.g., tocopheryl nicotinate, myristyl nicotinate) nicotinamide riboside, nicotinyl amino acids, nicotinyl alcohol esters of carboxylic acids, nicotinic acid N-oxide, and niacinamide N-oxide. In some instances, vitamin B₃ compounds such as niacinamide may have improved efficacy at lower pH, for example, as described in U.S. Publication No. 2020/0009123.

(47) In some instances, it may be desirable for the ring nitrogen of the vitamin B₃ compound to be "uncomplexed" (e.g., chemically unbound and/or unhindered) in the composition and/or prior to application to a target skin surface. For example, the compositions herein may be free of or substantially free of (i.e., less than 3%, 2%, 1% or even less than 0.5%) a salt or complex of a vitamin B₃ compound. Exemplary approaches to minimizing or preventing the formation of

undesirable salts and/or complexes include omission of materials that form substantially irreversible or other undesirable complexes with the vitamin B.sub.3 compound in the composition, pH adjustment, ionic strength adjustment, the use of surfactants, and practicing formulation processes wherein the vitamin B.sub.3 compound and materials which complex therewith are in different phases.

(48) Antioxidant

(49) The low-pH composition herein may include an antioxidant to combat HCA oxidation and/or degradation. The antioxidant, when included, may be present at 0.001% to 3% (e.g., 0.01% to 2%, 0.05% to 1%, or 0.1% to 0.5%). Some non-limiting examples of antioxidants that may be suitable for use herein are sodium sulfite, sodium bisulfite sodium metabisulfite, and butylated hydroxytoluene.

(50) Low-pH Acid Buffering System

(51) When providing a low-pH composition for topical application to skin, it can be important to include a buffering system to help maintain the pH of the composition for a period of time after it is applied to the skin (e.g., up to 5 minutes or more). On average, human skin pH typically ranges from about 5.0 to 6.0. To maintain this pH, human skin has evolved a natural buffering system that resists changes to pH. Thus, when a low-pH composition is applied to the skin, the skin's natural buffering system will try to adjust the pH of the composition to match the natural pH of the skin. Without the addition of the buffering agent, the low-pH composition may not be able to provide the desired skin care benefit. Accordingly, the compositions herein may include a low-pH acid buffering system.

(52) The buffering agent may be selected according to the acid(s) that is used to lower the pH of the low-pH compositions herein. For example, lactic acid and gluconic acid may be used, individually or in combination, to lower the pH of the composition because they are generally considered gentler on skin (i.e., lower risk of irritation) compared to other alpha hydroxy acids. In this example, sodium lactate and/or sodium gluconate would then be selected as the corresponding salt buffering agent to provide the acid/salt pH buffer system. The buffering agent may be present in the low-pH composition at 0.25% to 4% (e.g., 0.5% to 3%, 0.75% to 2% or 1% to 1.75%). A non-limiting example of a suitable low-pH buffer system for use herein is disclosed in co-pending U.S. Ser. No. 16/891,491. Of course, it is to be appreciated that the present composition may optionally include other pH buffers known for use in skin care compositions.

(53) Thickener

(54) The composition includes a polymer thickener that can tolerate a low-pH, electrolytic environment. That is, the thickener will not lose its ability to thicken or stabilize the composition at low-pH in the presence of an acid-salt buffering system. Some conventional neutralized thickeners are known to degrade and/or lose the ability to suitably thicken a composition at lower pH and/or in the presence of an acid-salt buffer (e.g., sodium lactate). For example, some neutralized thickeners degrade in a low-pH environment. On the other hand, fatty alcohol thickeners such as cetyl alcohols and stearyl alcohols are generally stable at low pH, but tend to impart an undesirable cloudiness or opacity to the composition when it is in the form of an essence, serum, or the like. It has also been found that certain anionic polymeric thickeners can provide suitable tolerance to low-pH environments but cannot tolerate buffer systems due to combination of acid and salt. Thus, in some instances, the low-pH composition described herein may be free or substantially free of neutralized thickeners, fatty alcohol thickeners, and anionic thickeners. The thickener may be present at 0.0001% to 25% (e.g., 0.001% to 20%, 0.01% to 10%, 0.5% to 7%, or 1% or 5%) by weight of the composition.

(55) Other nonlimiting examples of thickeners or water structuring agents that may be used alone or in combination herein include natural or synthetic gums, polysaccharides, carboxylic acid polymers, polyacrylamide polymers, sulfonated polymers, and copolymers of these. Further examples include modified gums, celluloses, and superabsorbent polymers. The term

“superabsorbent polymer” is understood to mean a polymer which is capable, in its dry state, of spontaneously absorbing at least 20 times its own weight of aqueous fluid, in particular of water and especially of distilled water. Suitable polysaccharides include alkyl hydroxyalkyl cellulose ethers, such as hydroxypropylmethylcellulose stearoxy ether. This material is sold under the tradename of SANGELOSE 60L and 90L from Daido Chemical Corp. Another suitable polysaccharide includes hydrophobically modified starch, such as Potato modified starch. This material is sold under the tradename of STRUCTURE SOLANACE by Nouryon. Another polymer includes crosslinked polymers, the monomers of which are at least partially composed of acryloyldimethyltaurate monomers, such as, for example sodium polyacryloyldimethyl taurate, sold under the tradename of ARISTOFLEX SILK, from Clariant.

(56) It has now been found that certain anionic polymeric thickeners can provide suitable tolerance to low-pH environments and the desired feel and opacity properties to the composition. Thus, a particularly suitable example of an anionic thickener is polyacrylate crosspolymer-6, which is commercially available as SEPIMAX ZEN from Seppic, France.

(57) Low Molecular Weight Silicone Fluid.

(58) In some instances, an anionic polymeric thickener may impart an undesirable tacky feel when the low-pH composition is applied to a target portion of skin. It has been found that the addition of a low molecular weight silicone fluid can reduce or prevent this tacky feel. The molecular weight of a silicone fluid depends on the length of its silicone polymer chain(s), which is also directly proportional to the viscosity of the silicone fluid. Thus, the low molecular weight silicone fluids suitable for use in the present low-pH composition have a kinematic viscosity of 100 cSt or less at 25° C. (e.g., 1 cSt to 90 cSt, 5 cSt to 50 cSt, or even 10 cSt to 30 cSt). Kinematic viscosity is a common method of classifying silicone fluids and can be obtained from the supplier of the material. A particularly suitable example of a low molecular weight silicone fluid is 5 cSt dimethicone fluid. As used herein, “dimethicone” means a polydimethylsiloxane compound having the formula:

(59) ##STR00003##

Dermatologically Acceptable Carrier

(60) The low-pH compositions herein include a dermatologically acceptable carrier (which may be referred to as a “carrier”). The phrase “dermatologically acceptable carrier” means that the carrier is suitable for topical application to the keratinous tissue, has good aesthetic properties, is compatible with the actives in the composition, and will not cause any unreasonable safety or toxicity concerns. In one embodiment, the carrier is present at a level of from about 50% to about 99%, about 60% to about 98%, about 70% to about 98%, or, alternatively, from about 80% to about 95%, by weight of the composition.

(61) The carrier can be in a wide variety of forms. In some instances, the solubility or dispersibility of the components (e.g., extracts, sunscreen active, additional components) may dictate the form and character of the carrier. Non-limiting examples include simple solutions (e.g., aqueous or anhydrous), dispersions, emulsions, and solid forms (e.g., gels, sticks, flowable solids, or amorphous materials). In some instances, the dermatologically acceptable carrier is in the form of an emulsion. The emulsion may have a continuous aqueous phase (e.g., an oil-in-water or water-in-oil-in-water emulsion) or a continuous oil phase (e.g., water-in-oil or oil-in-water-in-oil emulsion). The oil phase of the present invention may comprise silicone oils, non-silicone oils such as hydrocarbon oils, esters, ethers, and mixtures thereof. The aqueous phase typically comprises water and water-soluble ingredients (e.g., water-soluble moisturizing agents, conditioning agents, anti-microbials, humectants and/or other skin care actives). However, in some instances, the aqueous phase may comprise components other than water, including but not limited to water-soluble moisturizing agents, conditioning agents, anti-microbials, humectants and/or other water-soluble skin care actives. In some instances, the non-water component of the composition comprises a humectant such as glycerin and/or other polyol(s).

(62) In some instances, the compositions herein are in the form of an oil-in-water ("O/W") emulsion that provides a sensorial feel that is light and non-greasy. Suitable O/W emulsions herein may include a continuous aqueous phase of more than 50% by weight of the composition, and the remainder being the dispersed oil phase. The aqueous phase may include 1% to 99% water, based on the weight of the aqueous phase, along with any water soluble and/or water miscible ingredients. In these instances, the dispersed oil phase will typically be present at less than 30% by weight of composition (e.g., 1% to 20%, 2% to 15%, 3% to 12%, 4% to 10%, or even 5% to 8%) to help avoid some of the undesirable feel effects of oily compositions. The oil phase may include one or more volatile and/or non-volatile oils (e.g., botanical oils, silicone oils, and/or hydrocarbon oils). Some nonlimiting examples of oils that may be suitable for use in the present compositions are disclosed in U.S. Pat. No. 9,446,265 and U.S. Publication No. 2015/0196464.

(63) The carrier may contain one or more dermatologically acceptable, hydrophilic diluents. As used herein, "diluent" includes materials in which the vitamin B.sub.3 compound can be dispersed, dissolved, or otherwise incorporated. Hydrophilic diluents include water, organic hydrophilic diluents such as lower monovalent alcohols (e.g., C.sub.1-C.sub.4) and low molecular weight glycols and polyols, including propylene glycol, polyethylene glycol (e.g., molecular weight of 200 to 600 g/mole), polypropylene glycol (e.g., molecular weight of 425 to 2025 g/mole), glycerol, butylene glycol, 1,2,4-butanetriol, sorbitol esters, 1,2,6-hexanetriol, ethanol, isopropanol, sorbitol esters, butanediol, ether propanol, ethoxylated ethers, propoxylated ethers and combinations thereof.

(64) Emulsifier

(65) When the low-pH composition herein is in the form of an emulsion (e.g., oil-in-water emulsion), it may be desirable to include an emulsifier to stabilize the emulsion (i.e., prevent the emulsion from phase separating). The emulsifier may be present in the composition at 0.01% to 10% (e.g., 0.05% to 5% or 0.1% to 2%). The emulsifiers may be nonionic, anionic or cationic. In some instances, the emulsifier may be a silicone emulsifier. Some non-limiting examples of emulsifiers that may be suitable for use herein are disclosed in U.S. Pat. Nos. 3,755,560; 4,421,769; and McCutcheon's Detergents and Emulsifiers, North American Edition, pages 317-324 (1986).

(66) Some other non-limiting examples of emulsifiers that may be suitable for use herein include ethers of polyglycols and of fatty alcohols, esters of polyglycols and of fatty acids, ethers of polyglycols and of fatty alcohols which are glycosylated, esters of polyglycols and of fatty acids which are glycosylated, ethers of C12-30 alcohols and of glycerol or of polyglycerol, esters of C12-30 fatty acids and of glycerol or of polyglycerol, ethers of oxyalkylene-modified C12-30 alcohols and of glycerol or polyglycerol, ethers of C1-230 fatty alcohols comprising and of sucrose or of glucose, esters of sucrose and of C12-30 fatty acids, esters of pentaerythritol and of C12-30 fatty acids, esters of sorbitol and/or of sorbitan and of C12-30 fatty acids, ethers of sorbitol and/or of sorbitan and of alkoxylated sorbitan, ethers of polyglycols and of cholesterol, esters of C12-30 fatty acids and of alkoxylated ethers of sorbitol and/or sorbitan, and combinations thereof. A particularly useful class of emulsifiers is polyethylene glycol ethers of lauryl alcohol such as laureth-1 through laureth-50 (e.g., laureth-4). Still other examples of emulsifiers include ethers of glycerol, polyglycerol, sucrose, glucose, or sorbitol; esters of glycerol, polyglycerol, sucrose, glucose, or sorbitol; and mixtures thereof. Other particularly useful classes of emulsifiers are the alkyl esters of sorbitol and sorbitol anhydrides such as polysorbate 20, polysorbate 21, and polysorbate 40.

(67) In some aspects, it may be desirable to include a linear or branched silicone emulsifier in the low-pH composition. Particularly useful silicone emulsifiers may include polyether modified silicones such as KF-6011, KF-6012, KF-6013, KF-6015, KF-6015, KF-6017, KF-6043, KF-6028, and KF-6038 and polyglycerolated linear or branched siloxane emulsifiers such as KF-6100, KF-6104, and KF-6105; all from Shin-Etsu. A particular suitable emulsifier for use herein is PEG-11 methyl ether dimethicone, which is available from Shin-Etsu as KF-6011. It was discovered that the PEG-11 methyl ether dimethicone emulsifier further reduced the tacky feel of the anionic

polymer thickener, thereby improving the overall feel of the low-pH composition. The emulsifier may be present at an amount of 0.1% to 10% (e.g., 1% to 5%, or 2%-4%).

(68) Other Optional Ingredients

(69) The present composition may optionally include one or more additional ingredients commonly used in cosmetic compositions (e.g., colorants, skin care actives, anti-inflammatory agents, sunscreen agents, emulsifiers, buffers, rheology modifiers, combinations of these and the like), provided that the additional ingredients do not undesirably alter the skin health or appearance benefits provided by the present compositions. The additional ingredients, when incorporated into the composition, should be suitable for use in contact with human skin tissue without undue toxicity, incompatibility, instability, allergic response, and the like. Some nonlimiting examples of additional actives include vitamins, minerals, peptides and peptide derivatives, sugar amines, sunscreens, oil control agents, particulates, flavonoid compounds, hair growth regulators, antioxidants and/or anti-oxidant precursors, preservatives, protease inhibitors, tyrosinase inhibitors, anti-inflammatory agents, moisturizing agents, exfoliating agents, skin lightening agents, sunless tanning agents, lubricants, anti-acne actives, anti-cellulite actives, chelating agents, anti-wrinkle actives, anti-atrophy actives, phytosterols and/or plant hormones, N-acyl amino acid compounds, antimicrobials, and antifungals. Other non-limiting examples of additional ingredients and/or skin care actives that may be suitable for use herein are described in U.S. Publication Nos.

2002/0022040; 2003/0049212; 2004/0175347; 2006/0275237; 2007/0196344; 2008/0181956; 2008/0206373; 2010/00092408; 2008/0206373; 2010/0239510; 2010/0189669; 2010/0272667; 2011/0262025; 2011/0097286; US2012/0197016; 2012/0128683; 2012/0148515; 2012/0156146; and 2013/0022557; and U.S. Pat. Nos. 5,939,082; 5,872,112; 6,492,326; 6,696,049; 6,524,598; 5,972,359; and 6,174,533.

(70) When including optional ingredients in the compositions herein, it may be desirable to select ingredients that do not form complexes or otherwise undesirably interact with other ingredients in the composition at low-pH, especially pH sensitive ingredients like niacinamide, salicylates and peptides. In some instances, it may be desirable to select skin care actives that function via different biological pathways so that the actives do not interfere with one another, which could reduce the efficacy of both agents. When present, the optional ingredients may be included at amounts of from 0.0001% to 50%; from 0.001% to 20%; or even from 0.01% to 10% (e.g., 50%, 40%, 30%, 20%, 10%, 5%, 4%, 3%, 2%, 1%, 0.5% or 0.1%), by weight of the composition.

(71) Method of Use

(72) The low-pH compositions herein are formulated for topical application to skin. The method of using the present low-pH composition involves identifying a target portion of skin on a person in need of treatment or where treatment is desired (e.g., portions of skin exhibiting uneven tone, sallowness, hyperpigmented spots, fine lines or wrinkles, skin stability) and applying an effective amount of the low-pH composition to the target portion of skin over the course of a treatment period. The effective amount of composition may vary based on the skin benefit desired by the user and/or the size of the treatment area. In some instances, the effective amount may range from 0.1 g to 5 g (e.g., 0.2 g to 4 g, 0.3 g to 2 g, or even 0.5 g to 1 g). The target portion of skin may be on a facial skin surface such as the forehead, perioral, chin, periorbital, nose, and/or cheek) or another part of the body (e.g., hands, arms, legs, back, chest). In some instances, a target portion of skin may be selected that does not currently exhibit signs of skin aging, such as hyperpigmented spots or uneven skin tone, but is an area of skin that commonly exhibits such features with age. In these instances, the low-pH composition may be used to help prevent the occurrence of such undesirable skin features.

(73) The composition may be applied locally to the target portion of skin in need of treatment and, if desired, to the surrounding skin at least once a day, twice a day, or on a more frequent daily basis, during a treatment period. When applied twice daily, the first and second applications are separated by at least 1 to 12 hours. Typically, the composition is applied in the morning and/or in the evening

before bed. When used according to the methods herein, the present compositions may improve the appearance and/or function of skin, for example, by improving skin texture. Improvements in skin texture can be provided, for example, by decreasing pore size, reducing skin roughness, reducing the presence and/or size of wrinkles, combinations of these and the like.

(74) The treatment period is ideally of sufficient time for the low-pH composition to improve the appearance and/or function of the target portion of skin. The treatment period typically lasts for at least 1 week (e.g., about 2 weeks, 4 weeks, 8 weeks, or even 12 weeks). In some instances, the treatment period may extend over multiple months (i.e., 3-12 months). In some instances, the composition is applied most days of the week (e.g., at least 4, 5 or 6 days a week), at least once a day or even twice a day during a treatment period of at least 2 weeks, 4 weeks, 8 weeks, or 12 weeks.

(75) The step of applying the composition herein may be accomplished by localized application. In reference to application of the composition, the terms “localized”, “local”, or “locally” mean that the composition is delivered to the targeted area (e.g., a psoriatic plaque) while minimizing delivery to skin surfaces where treatment is not desired. The composition may be applied and lightly massaged into an area of skin. The form of the composition or the dermatologically acceptable carrier should be selected to facilitate localized application. While certain embodiments herein contemplate applying a composition locally to an area, it will be appreciated that compositions herein can be applied more generally or broadly to one or more skin surfaces. In certain embodiments, the compositions herein may be used as part of a multi-step beauty regimen, wherein the present composition may be applied before and/or after one or more other compositions.

Methods

(76) HPLC

(77) This method provides a way to determine the weight percentage of HCA and 4-vinylphenol (4-VP), respectively, in raw materials or finished products using high performance liquid chromatography (“HPLC”). This method may also be used to identify HCA and/or 4VP, by matching their wavelength spectrum and retention time to their respective known standards. The following instruments and materials are used in this method:

(78) Instruments:

(79) a gradient HPLC system that includes a gradient HPLC pump, liquid auto-sampler, UV detector (and diode array detector (DAD) for spectrum analysis), and a suitable computing integrator or computer data system (e.g., a Waters 2695 HPLC system from Waters Corporation or equivalent). a 5 μ m, 250 mm $\times$ 4.6 mm ID HPLC column (e.g., a C18 column from Alltech Alltima).

Method:

(80) Two mobile phases are used to create the gradient with a consistent flow rate of 1.0 mL/min. Mobile phase A consists of 0.5% acetic acid in purified water. Mobile phase B consists of 0.5% acetic acid in acetonitrile. The gradient is illustrated below in Table 1.

(81) TABLE-US-00001 TABLE 1 Analysis Mobile Phase Composition (v/v) Time %A %B 0.00 90 10 6.00 90 10 7.00 50 50 12.00 50 50 13.00 30 70 16.00 30 70

(82) The column temperature is 25° C. Data collection UV wavelength is 263 nm. The spectrum data between 190 nm and 400 nm are collected to identify the HCA and 4-VP. The retention time for HCA is 10.86 ($\pm$ 0.20) minutes, and the retention time for 4-VP is 14.60 ($\pm$ 0.20) minutes. The reference spectrum for HCA and 4-VP are shown in FIGS. 1A and 1B, respectively.

(83) Calculations

(84) 1. Calculation of wt % of HCA in finished product samples

(85) $HCA_{Wt\%} = \frac{A \times M \times P \times 100}{B \times W}$ A=Peak area ratio of HCA:Internal Standard in sample B=Peak Area ratio of HCA:Internal Standard in Calibration Standard M=Mass of HCA in mg, in Calibration Standard (~50 mg/50 mL $\times$ 3 mL) W=Sample weight in mg P=Purity of HCA in decimal 2.

Calculation of wt % of 4-VP in Finished Product Samples

(86) $4VPWt\% = \frac{a \times m \times p \times 100}{b \times w}$ a=Peak area ratio of 4-VP:Internal Standard in sample b=Peak area ratio of 4-VP:Internal Standard in Calibration Standard m=Mass of 4-VP in mg, in Calibration Standard (~150 mg/100 mL×1 mL) W=Sample weight in mg p=Purity of 4-VP in decimal

HCA Crystallization

(87) This method provides a way to determine HCA solubility in a composition by observing HCA crystals in situ. The method involves cycling the temperature of a test sample between freezing and thawing to imitate environmental conditions experienced by a skin care composition at an accelerated rate. This type of accelerated aging is commonly used in cosmetic product stability testing. HCA crystals can be detected using conventional means such as visual observation and microscopy.

(88) A bulk sample of at least 10 g (e.g., 20 g-60 g) of the composition to be tested is placed in a suitable container that enables visual observation of the test sample (e.g., transparent plastic or glass jar). The test sample is subjected to 1 month of freeze/thaw temperature cycling to simulate environmental conditions that a skin care product may experience during shipping and storage. This is sometimes referred to as accelerated aging. The temperature cycling involves a 1-week freeze cycle at -7°C . (or -18°C ., if noted), followed by a 1-week thaw cycle at 25°C ., and then repeating this freeze/cycle for a total temperature cycling time of 1 month.

(89) Upon completion of the accelerated aging process (i.e., 1 month of temperature cycling), transparent test samples are visually inspected in-situ in the transparent container to determine if HCA crystallization/precipitation occurred. For opaque and translucent samples, the entire test sample is removed from the container and transferred to a suitable transparent substrate (e.g., plastic film or glass plate) and formed into a thin film of no more than 1 mm thickness. The sample is covered with a second transparent substrate to inhibit the loss of volatile ingredients during inspection. A light source (e.g., LED lamp or the like) is used to backlight the sample to aid in visual inspection. HCA crystals will generally appear as a precipitate in the composition visible to the naked eye when observed by someone with 20/20 vision from 45 cm away. Any precipitate identified during visual observation may be further evaluated using a microscope capable of providing fluorescent birefringence observation with cross-polarized light to identify anisotropic crystals. Any anisotropic crystal that has a longest dimension of greater than 0.1 mm is identified as an HCA crystal and the total number of HCA crystals is recorded. A test sample that contains no more than 1 HCA crystal is considered to be “free of HCA crystals” and is recorded as a “pass.” A test sample that contains more than 1 HCA crystal, is recorded as a “fail.”

(90) While not required, fourier-transform infrared spectroscopy (FTIR) can be used to confirm that the HCA crystal or co-crystal contains the appropriate hydroxycinnamic acid structure. FTIR spectroscopy techniques are well-known in the art. See, U.S. Pat. No. 10,912,857, US2020/0000697, and Fourier Transform Infrared Spectroscopy in Colloid and Interface Science, D. R. Scheuing, Ed., American Chemical Society, 225, 1991.

(91) Saturated Solubility

(92) This method provides a way to determine the saturated solubility of HCA compositions, using UV-Vis-Spectroscopy.

(93) Instruments and Materials:

(94) A UV-VIS-NIR Spectrophotometer (such as UV-3600 Shimadzu UV-VIS-NIR Spectrophotometer) with computer data system. A suitable sample cell (such as the 1 cm path length quartz cells). A centrifuge, Srvall ST40R Centrifuge, and syringe filter (0.2 μm syringe filter, H-PTFE, 13 mm). A Thermo Scientific™ LP Vortex Mixer and 70% ethanol acidic solution (69.3% ethanol, 29.7% water, 1% HCl), pH<3.

(95) Saturated Solubility Sample Preparation:

(96) The saturated solubility is measured by preparing a 100 ml samples in glass beaker with HCA added in excess. Adjust the solution pH to target using HCl/NaOH solution. Leave the solution mixing overnight to reach equilibrium. Aliquot 20 ml solution into a 50 ml centrifuge tube and

centrifuge at 4700 rpm, 10 mins, 25° C. Filter the required sample amount through the 0.2 um syringe filter. Dilute solution using 70% ethanol acidic solution to desired dilution factor.

(97) Oil-Water Partitioning Sample Preparation:

(98) The measurement is similar to the saturated solubility test but is used to measure the relative partitioning between the oil and water phases of a multiple phase formulation. In this case, prepare 20 ml samples in 50 ml centrifuge tube. Mix samples for 10 seconds using the vortex mixer. Leave the sample to equilibrate overnight. Centrifuge samples at 4700 rpm, 10 mins, 25° C. After centrifugation, clear oil and water phases should be visibly distinct. Aliquot needed oil phase and dilute to desired dilution factor using 70% ethanol acidic solution. Aliquot a separate needed water phase, filter through the 0.2 um syringe filter, and dilute to desired dilution factor using 70% ethanol acidic solution.

(99) Measurement:

(100) Measurements are done per UV-Vis Spectrophotometer manufacturer guidelines. Turn on UV machine and warm up the lamp for 20 mins, ensure calibrations and standard dilution curves are generated. For HCA, set the wavelength to 200-500 nm, scan speed: fast, single scan mode with auto sampling interval, and absorbance mode. Measure the background using 2× quartz cells with 70% ethanol acidic solution. Measure sample and read the absorbance at 311 nm. Calculate the measured HCA concentration from the absorbance at 311 nm vs. the calibration curve and adjust to the final concentration based on the dilution factor.

(101) Calculations

(102) 1. Calculation of measured HCA concentration in water or oil phase

(103) $\text{Concentration HCA} \left(\frac{\text{mg}}{\text{ml}} \right) = \frac{\text{peak at 311 nm} \times \text{dilution factor}}{\text{gradient of calibration curve} \times 1000}$ 2. Calculation of measured HCA weight %

(104) $\text{HCA (wt\%)} = \frac{\text{measured HCA (g)}}{\text{quantity of HCA added (g)}} \text{HCA (g)} = \frac{\text{HCA (ppm)} \times \text{volume (ml)} \times \text{density} \left(\frac{\text{g}}{\text{ml}} \right)}{1000000}$

$\text{HCA (ppm)} = \frac{\text{peak at 311 nm} \times \text{dilution factor}}{\text{gradient of calibration curve}}$ Gradient of calibration curve (70% ethanol acidic

solution)=0.1229 Volume (ml)=total amount of measured phase (oil or water), excluding solutes. Density of water and oil phase is assumed to be 1 g/ml.

Color

(105) This method can be used to determine the change in color of a product, material, or target portion of skin. A spectrophotometer (e.g., Spectrophotometer CM-3600A, Konica Minolta, Japan, or equivalent) is used to measure spectrometric data for CIE (International Commission on Illumination) specified illuminating and viewing conditions, and compute the associated tristimulus values, XYZ, based on CIE observer and a CIE illuminant standards. The ASTM standard for obtaining spectrometric data for Object-Color Evaluation is ASTM E1164-12(2017)e1 with the values and procedures for computing CIE tristimulus values from spectrometric data outlined in ASTM E308. The CIELAB color scale, also referred to as L*a*b*, are calculated from the tristimulus values as defined in ASTM E308. The L* denotes the sample's perceptual lightness. Whereas a* and b* relate to the unique colors of human vision, with a* being positive in the red direction and negative in the green direction, and b* positive in the yellow direction and negative in the blue direction.

(106) The spectrophotometer is operated with a 2° observer and D65 illuminant to measure 1931 CIE defined tristimulus XYZ values with and associated CIELAB colors. Portion samples are evaluated using a 10 mm path length in a plastic cell (e.g., CM-A131, Konica Minolta, Japan or equivalent), reflectance measurement, a 25.4 mm aperture opening at the specimen surface, and specular component excluded conditions with a standard white background. Acceptable white backgrounds include the white portion of an opacity card or equivalent (e.g., opacity card Form 2A, Leneta Company, Inc, Mahwah, N.J.). The instrument is calibrated with a zero standard, utilizing a zero-calibration box, or black trap, that is free from dust and debris (e.g., Konica Minolta zero calibration box CM-A155 or equivalent). The CIELAB values are reported by the associated

instrument software (e.g., SpectroMagic NX) as defined in ASTM E308. For changes in product yellowness, the CIELAB color scale is utilized focused on the change in the b* value. The results are reported as a Δb^* , where a positive Δb^* indicates increased yellowness.

(107) Rheology

(108) This method provides a way to measure the dynamic viscosity of a composition or material using a BROOKFIELD brand viscometer (e.g., model DV2T or equivalent) and a suitable spindle (e.g., RV4 or equivalent) according to the manufacturer's instructions. It is to be appreciated that the skilled artisan will be able to select the appropriate spindle in accordance with the manufacturer's recommendation. After calibrating the viscometer, the spindle is immersed into a sufficient quantity of test sample (e.g., enough to immerse the spindle up to the immersion mark on the spindle shaft). Set the spindle rotation speed to 5 rpm, and then start the viscometer. Allow time for the indicated viscosity reading to stabilize (approximately 10-30 seconds). After the reading stabilizes, take 5 readings at 10 second intervals. Calculate the viscosity as the average of the 5 readings.

EXAMPLES

Example 1: Formulations

(109) Table 2 below provides examples of low-pH skin care compositions that correspond to various aspects of the invention. The compositions can be prepared using conventional methods of making skin care compositions. Such methods typically involve mixing of the ingredients in one or more steps to a relatively uniform state, with or without heating, cooling, application of vacuum, and the like. All exemplified amounts exclude minor materials such as diluents, preservatives, color solutions, feel modifying powders and elastomers, etc., that may be present in a commercial product unless otherwise specified. HCA may be added as a solid form and solubilized in-situ, dissolved as a premix, or supplied as a pre-dispersed raw material. A pre-dispersed 15% solution of 4-HCA in PEG-4 (Lipobrite® from Vantage) is used for certain examples. For examples containing Lipobrite®, the 4-HCA and PEG-4 constituent levels are listed individually for clarity, with a superscript to denote the Lipobrite® raw material. The total Lipobrite® material added is the sum of the 4-HCA and PEG-4 constituents listed. All other materials are listed 'as is' from the suppliers.

(110) Emulsions are prepared by first mixing the aqueous phase materials separately from the oil and/or silicone phase materials and then combining the two phases as appropriate to yield the desired continuous phase. In some aspects, the exemplary compositions can be made by blending the aqueous phase components with a suitable mixer (e.g., IKA RW20 or equivalent) until all materials are dissolved and homogeneous. When present, the optional polymer thickener may be hydrated by slowing adding the thickener directly into a water phase while stirring and continued mixing until homogeneous. The 4-HCA, polar emollient, and glycol co-solvent may be added together in a separate pre-mix container and mixed until fully dissolved and uniform. The HCA premix can then be added to the main mix container and further mixed until homogeneous. The formula can be milled using a suitable mixer (e.g., IKA Ultra Turrax T-25 or equivalent) to reduce the emulsion particle size until a target viscosity is reached and uniform composition is achieved. The temperature may be adjusted as needed to control the speed of the process and/or achieve a homogenous final product.

(111) TABLE-US-00002	TABLE 2	A	B	C	D	E	F	G	Component	%	Water	qs	qs	qs	qs	qs	qs	qs
Glycerin	7.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	Dimethicone 5 cSt	—	—	2.00	2.00	2.00	2.00	2.00	2.00	2.00
Niacinamide	2.00	2.00	2.00	2.00	2.00	2.00	5.00	2.00	Lactic acid	1.80	1.80	1.80	1.80	1.80	1.80	1.80	1.80	1.80
Sodium lactate	1.30	1.30	1.30	1.30	1.30	1.30	1.30	1.30	Polyacrylate crosspolymer-6	.sup.1	—	—	—	—	—	—	1.25	1.25
1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25	Sodium polyacryloyldimethyl taurate	.sup.2	1.25	1.25	—	—	—	—	—	D-
Panthenol	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	Disodium EDTA	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10
PEG-11 methyl ether dimethicone	.sup.3	—	—	—	—	—	—	—	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10
0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	Trehalose	—	—	—	—	—	—	—	0.50	0.50
0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	Sodium benzoate	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	Sodium sulfite	—	—	—	—	—	—	—	0.05	0.05
0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	4-HCA .sup.4	0.0001	1.00	0.50	0.25	0.10	1.00	0.30	PEG-4 .sup.4	0.00057
0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.00057	5.67	—	—	—	—	—	—	—	—

2.83 1.42 0.57 5.67 1.7 Isopropyl lauroyl sarcosinate .sup.5 0.10 5.00 4.00 3.00 1.00 5.00 1.00
Pentylene glycol 0.10 3.00 3.00 3.00 3.00 3.00 3.00 Propylene glycol — 5.00 — — — 5.00 5.00
Dipropylene glycol — 4.00 — — — 4.00 4.00 Ethoxydiglycol — 2.60 — — — 2.60 —
Isohexadecane 3.00 3.00 — — — — — Isopropyl isostearate 1.50 1.50 — — — — — Cetearyl
glucoside and cetearyl 0.50 0.50 — — — — — Behenyl Alcohol 0.70 0.70 — — — — — Stearyl
Alcohol 1.20 1.20 — — — — — Cetyl Alcohol 0.90 0.90 — — — — — PEG-100 Stearate 0.10
0.10 — — — — — Stearic Acid 0.10 0.10 — — — — — Dimethicone and dimethiconol .sup.6
1.00 1.00 — — — — — Palmitoyl pentapeptide-4 .sup.7 0.50 0.50 — — — — — NaOH * * * * *
* * HCl * * * * * pH 4.0 4.0 4.0 4.0 3.0 5.0 3.8 H I J K L M N Ingredient % Water qs qs qs qs
qs qs qs Glycerin 3.00 3.00 3.00 7.00 3.00 3.00 3.00 Dimethicone 5 cSt 2.00 2.00 — — 2.00 2.00
2.00 Niacinamide 2.00 0.50 2.00 0.10 0.50 0.10 2.00 Lactic acid 1.80 1.80 1.80 1.80 1.80 1.80 1.80
Sodium lactate 1.30 1.30 1.30 1.30 1.30 1.30 1.30 Polyacrylate crosspolymer-6 .sup.1 1.20 1.20
1.20 1.25 1.20 1.20 1.20 Sodium polyacryloyldimethyl taurate .sup.2 — — — — — — —
Panthenol 0.50 0.50 0.50 0.50 0.50 0.50 0.50 Disodium EDTA 0.10 0.10 0.10 0.10 0.10 0.10 0.10
PEG-11 methyl ether dimethicone .sup.3 0.10 0.10 0.10 — 0.10 0.10 0.10 Trehalose 0.10 0.10 0.10
0.10 0.10 0.10 0.10 Sodium benzoate 0.05 0.05 0.05 0.05 0.05 0.05 0.05 Ferulic acid 0.25 0.25 0.25
— — — — 4-HCA .sup.4 0.25 0.25 0.25 0.10 0.5 0.5 0.5 PEG-4 .sup.4 1.42 1.42 1.42 0.57 2.83
2.83 2.83 Isopropyl lauroyl sarcosinate .sup.5 4.00 4.00 4.00 1.00 1.00 3.00 1.00 Pentylene glycol
3.00 3.00 — 3.00 3.00 3.00 3.00 Propylene glycol — — — — 5.00 5.00 5.00 Dipropylene glycol
— — — — 4.00 4.00 4.00 Isohexadecane — — 3.00 3.00 — — — — Isopropyl isostearate — —
1.50 1.50 — — — Cetearyl glucoside and cetearyl — — 0.50 0.50 — — — Behenyl alcohol — —
0.70 0.70 — — — Stearyl alcohol — — 1.20 1.20 — — — Cetyl alcohol — — 0.90 0.90 — — —
PEG-100 stearate — — 0.10 0.10 — — — Stearic acid — — 0.10 0.10 — — — Dimethicone and
dimethiconol .sup.6 — — 1.00 1.00 1.00 — — Palmitoyl pentapeptide-4 .sup.7 — — 0.5 0.50 —
— — NaOH * * * * * HCl * * * * * pH 4.0 5.0 4.0 2.5 3.0 2.5 4.0 .sup.1 SEPIMAX ZEN
available from Seppic .sup.2 ARISTOFLEX SILK available from Clariant .sup.3 KF-6011
available from Shin-Etsu .sup.4 LIPOBRITE from Vantage (15% 4-HCA, 85% PEG-4) .sup.5
Eldew SL-205 from Ajinomoto OmniChem .sup.6 DC-1503 from Dow Corning .sup.7 Ferulic Acid
from Sigma Aldrich .sup.8 PROMATRIXYL from Croda * pH adjustment as necessary

Example 2: Effect of pH and Niacinamide on HCA Stability

(112) This example demonstrates the effect of pH and niacinamide on HCA stability. When HCA degrades (e.g., via decarboxylation), it produces 4-vinylphenol as a byproduct, which is generally undesirable in a skin care composition. 4-VP can cause the composition to have an unpleasant odor and may further react to cause discoloration (e.g., yellowing) of the composition. Thus, it may be desirable to limit the amount of 4-VP in a skin care composition to less than 1500 ppm (e.g., less than 1100, 800 ppm, 700 ppm, 600 ppm, 500 ppm, 400 ppm, or even less than 300 ppm).

(113) The compositions shown in Table 3, were tested to determine the effect of pH and niacinamide on the degradation of HCA and the formation of 4-VP. The pH of the test compositions was adjusted using a 6N HCL solution or 4% NaOH solution, as necessary. The test compositions were incubated at 50° C. for 3 weeks, and then tested according to the HPLC method described above for HCA degradation and the color method described above for changes in color (Δb^*).

(114) TABLE-US-00003 TABLE 3 3A 3B 3C 3D 3E 3F Component % Water qs qs qs qs qs qs
HCA 1.0 1.0 1.0 1.0 1.0 1.0 Niacinamide 5 5 5 0 0 0 Pentylene Glycol 40 40 40 40 40 40 pH 3.56
4.57 6.17 3.45 4.52 6.10 Results (3W50C): HCA wt % 0.921 0.915 0.888 0.977 0.966 0.949
(remaining) 4-Vinylphenol 380 700 1070 90 200 490 Color Db* 3.51 0.95 6.04 0.43 0.45 1.6

(115) As can be seen in Table 3, both pH and niacinamide appear to affect HCA degradation, Δb^* and 4-VP formation. These data suggest that increasing pH and/or adding niacinamide will increase the degradation of 4-HCA, increase the formation of 4-VP, and cause yellowing in an aqueous skin care composition. The data unexpectedly show that adding niacinamide actually accelerates the

formation of 4-VP at higher pH. This is unexpected because it was previously unknown that niacinamide contributed to HCA degradation at all, let alone accelerated it. Thus, lowering the pH in a skin care composition comprising HCA and niacinamide appears to reduce the rate of formation of 4-VP more than expected.

(116) The data in Table 3 also demonstrate that both pH and niacinamide contribute to the change in yellowness (Δb^*) of the test compositions. Surprisingly, a pH of between 3.8 and 6.0 (e.g., about 4.5) appears to provide the lowest Δb^* for the test compositions containing niacinamide. This result is surprising because it was believed that yellowing (i.e., a positive change in Δb^* value) associated with HCA degradation would change with pH in a predictable linear fashion, as seen in with compositions 3D-3F.

Example 3: Effect of a Radical Scavenger on Rate of Yellowing

(117) This example demonstrates the ability of an antioxidant to help reduce HCA degradation as evidenced by a reduced rate of yellowing (i.e., lower Δb^* value). The test compositions shown in Table 4 below were tested with and without sodium metabisulfite, an oxygen-quenching, reducing agent, at pH 3.8 and pH 6.0. The test compositions were aged at two different accelerated aging conditions: 3 weeks at 50° C. (3W50C) and 3 months at 40° C. (3M40C). It is believed, without being limited by theory, that the accelerated aging conditions simulate environmental conditions that a skin care product may be exposed to during shipping and storage. Following the accelerated aging, the test compositions were tested according to the Color method described above to determine Δb^* . The test composition aged for 3 months at 40° C. were further tested according to the HPLC method to determine the level of HCA degradation and 4-VP formation.

(118) The results of the testing are summarized in Table 4 below. As can be seen in Table 4, the addition of sodium metabisulfite (SMBS) significantly reduced the rate of yellowing for compositions 4B and 4D, as compared to 4A and 4C, respectively. Surprisingly, the data indicate that the effect of the oxygen quenching reducing agent was enhanced more than expected at low pH vs. neutral pH (i.e., 20% for 3W50C and 240% for 3M40C, as can be seen in Table 5). The reduction in Δb^* does not appear to be due to a further reduction of HCA degradation or 4-VP formation, but some other unique mechanism of action. Thus, it is now possible to provide combinations of ingredients that can synergistically reduce the rate of yellowing in a low-pH composition containing HCA.

(119) TABLE-US-00004

TABLE 4 % Component	4A	4B	4C	4D	Water	qs	qs	qs	qs	Glycerin	4.5	4.5	
4.5 4.5 Dimethicone	5	cSt	4.00	4.00	4.00	4.00	Niacinamide	2.00	2.00	2.00	2.00	Lactic acid	2.00
2.00 — — Sodium lactate	0.90	0.90	—	—	Polyacrylate crosspolymer-6	sup.2	1.50	1.50	1.50	1.50	Panthenol	0.50	
0.50 0.50 0.50 0.50 PEG-11 methyl ether dimethicone	sup.4	0.10	0.10	0.10	0.10	Sodium Benzoate	0.05	0.05	0.05	0.05	4-HCA	sup.5	
1.00 1.00 1.00 1.00 PEG-4	sup.5	5.67	5.67	5.67	5.67	Ethoxydiglycol	20.0	20.0	20.0	20.0	Sodium Metabisulfite	—	
0.05 — 0.05 NaOH (adjust to target pH)	—	—	0.22	0.22	pH (target)	3.8	3.8	6	6	Results (3W50C)	Db* value	7.73	
2.13 10.73 6.04 Results (3M40C)	Db* value	8.42	2.11	11.35	8.71	HCA wt % remaining	0.891	0.891	0.907	0.882	4-VP	623	
638 1044 980													

(120) The synergistic results from the testing in this example are illustrated in Table 5. Synergy factor is calculated by dividing the observed value by the expected value. A synergy factor of greater than 1 indicates a synergistic effect.

(121) TABLE-US-00005

TABLE 5	3W50C	Db*	3M40C	Db*	Control [4C]	10.73	11.35	pH Effect,
pH 3.8 vs. pH 6 [4A – 4C]	–3.00	–2.93	SMBS Effect at pH 6, [4D – 4C]	–4.69	–2.64	4B Expected		
[4D + (4A – 4C)]	3.04	5.78	AND/OR [4B+ (4D – 4C)]	Expected combined effect [4B.sub.expected – 4C]	–7.69	–5.57	4B Actual	
2.13 2.11	Actual combined effect [4B.sub.actual – 4C]	–8.6	–9.24	Synergy Factor, [5B.sub.actual – 5C]/	1.12	1.66	[5B.sub.expected – 5C]	

Example 4: Low pH Solubility

(122) This example demonstrates the ability of the low-pH compositions herein to solubilize HCA at low pH. As illustrated in FIG. 2, HCA solubility begins to decrease dramatically at

1W60C. The increase in niacin formation from 1W60C accelerated aging is reported as the delta versus the initial condition tested to account for any differences in the time taken between testing the ambient retains. It is clearly shown that the inventive samples have significantly reduced niacin by an order of magnitude versus the comparative examples in the prior art which claimed to use niacinamide as a hydrotrope.

(127) TABLE-US-00008 TABLE 8 Niacin (wt %) Niacin (ppm) Composition Δ (wt %) 1W60C Δ (ppm) 1W60C Comparative-Table 7A 0.090 900 Comparative-Table 7B 0.135 1350 Comparative-Table 7C 0.142 1420 Comparative-Table 7D 0.114 1140 Comparative-Table 7E 0.127 1270 Comparative-Table 7F 0.160 1600 Comparative-Table 7G 0.158 1580 Comparative-Table 7I 0.163 1630 Inventive-Table 6A 0.0054 54 Inventive-Table 6B 0.0069 69

Example 6: HCA Saturated Solubility Measurements

(128) This example describes comparative and inventive compositions related to solubilizing HCA in low pH by measuring the aqueous saturated solubility and HCA concentrations in the oil and water phases. The saturated solubility of HCA in the aqueous phase of the compositions described in Table 9 were measured by the UV-VIS spectroscopy measurements detailed in the methods section. The test compositions shown in Table 9 were subjected to 2 freeze/thaw cycles at -18° C./25° C. for 1 week/1 week, totaling one month of temperature cycling. It has been found that the use of glycols alone does not sufficiently boost the solubility of HCA in low pH aqueous compositions resulting in crystallization. Furthermore, it has been found that traditional non-polar oils (e.g., isohexadecane) do not fully solubilize HCA in oil and water phase compositions. It was found that the addition of a polar oil alone (e.g., isopropyl lauroyl sarcosinate) is not sufficient to fully solubilize HCA in low pH compositions if the balance of polar oils and co-solvents (such as glycols) is not fully optimized such as 9A, 9C examples. It was also interestingly found that non-polar oils reduce the relative concentration of HCA in the oil phase even when an equivalent amount of the polar oil is present, reference 9C vs. 9D. As such, optimizing the oil-water partitioning along with boosting the aqueous phase saturated solubility is critical to solubilizing HCA in low pH compositions. Examples of comparative and inventive compositions are shown below:

(129) TABLE-US-00009 TABLE 9 Phase Material 9A 9B 9C 9D A p-Coumaric acid 0.3 0.3 0.3 A PEG-4 1.7 1.7 1.7 1.7 A Propylene Glycol — — 5 A Dipropylene Glycol — — — 4 A Pentylene Glycol 3 3 3 3 B Water QS QS QS QS B Nicotinamide 2 2 2 2 C Isopropyl lauroyl 1 3 1 1 sarcosinate .sup.1 C Isohexadecane — — 10 — D 10% NaOH * * * * (pH adjuster) D HCl (pH adjuster) * * * * Total 100 100 100 100 Target 3.8 3.8 3.8 3.8 pH Total (A)Aqueous Phase 1.53 0.61 1.37 1.65 measured concentration (mg/mL) (B)Aqueous Phase 0.99 0.99 0.99 1.77 measured Saturated Solubility (mg/mL) (C)Aqueous Phase 0.149 0.0579 0.119 0.160 measured HCA (wt %) (D)Oil Phase HCA (wt %) 0.131 0.236 0.00667 0.107 Freeze/Thaw Fail Pass Fail Pass (1 month-18/25 C.)

(130) As shown by the above table 9, a key criterion to ensure that compositions do not crystallize is to such that the measured initial concentration of HCA (row A) is less than the saturated solubility (row B). Oversaturated solutions may appear to be fully dissolved during making but will likely crystalize over time. Given the compositions are aqueous based, the saturated solubility is measured in the aqueous phase as detailed in the methods section. The HCA concentration should be measured promptly after dissolution and composition making to ensure that false positives are not observed. For example, if samples are not measured promptly within the same day, HCA may crystallize out of solution such that the measured initial aqueous concentration is falsely measured at equal to or less than the saturated solubility whereas the initial solution was oversaturated.

(131) It was found that an effective method to reduce the aqueous phase concentration less than the saturated solubility is to increase the oil-to-water partitioning (row C to row D). This is done by

increasing the HCA concentration in the oil phase versus the aqueous phase concentration. It was found that polar emollients/oils, such as isopropyl lauroyl sarcosinate, are more effective in boosting the oil phase concentration than other oil phase solvents. This example shows the effective combination of polar emollients with aqueous phase co-solvents to reduce crystallization of HCA at low pH.

Example Combinations

(132) 1. A low-pH, aqueous skin care composition, comprising: a. about 0.1% to about 10% of a hydroxycinnamic acid (HCA); b. about 0.1% to about 10% of a polar emollient; c. from about 1.5% to less than 20% of a co-solvent with a Hansen solubility parameter distance of less than 15 from hydroxycinnamic acid (HCA); and d. water, wherein the pH of the composition is about 5.0 or less, wherein the composition meets at least one of the following conditions (i)-(iii): (i) the composition further contains from about 0.1% to about 10% of a vitamin B3 compound (ii) the weight ratio between HCA and the polar emollient is from about 1:1 to about 1:10 (iii) the composition further contains from about 0.1% to 5% of a sensory improving ingredient selected from the group consisting of sugar alcohols, sugar-based emulsifiers, modified sugar derivatives, and mixtures thereof. 2. The composition of the preceding feature, wherein the sensory improving ingredient is preferably a sugar alcohol. 3. The composition of any of the preceding features, the weight ratio between sensory improving ingredient and co-solvent is from about 2:1 to about 1:20, 4. The composition of any of the preceding features, the weight ratio between sensory improving ingredient and co-solvent is from about 1.5:1 to about 1:10, 5. The composition of any of the preceding features, the weight ratio between sensory improving ingredient and co-solvent is from about 1.5:1 to about 1:5. 6. The composition of any of the preceding features, wherein the composition is free of HCA crystals. 7. The composition of any of the preceding features, further comprising a low-pH tolerant polymer thickener. 8. The composition of any of the preceding features, wherein the thickener is polyacrylate crosspolymer-6. 9. The composition of any of the preceding features, wherein the co-solvent is a glycol selected from the group consisting of propylene glycol, dipropylene glycol, butylene glycol, pentylene glycol, hexylene glycol, ethoxydiglycol, C2-C6 polyethene glycols, and combinations thereof. 10. The composition of any of the preceding features, further comprising about 0.001% to about 5% by weight of an antioxidant. 11. The composition of any of the preceding features, wherein the antioxidant is selected from the group consisting of sodium sulfite, sodium bisulfite, sodium metabisulfite, and combinations thereof. 12. The composition of any of the preceding features, wherein the pH is about 2.0 to about 4.5. 13. The composition of any of the preceding features, wherein the composition exhibits a Δb^* of less than about 6 according to the Color Method. 14. The composition of any of the preceding features, wherein the Δb^* is less than about 2.5. 15. The composition of any of the preceding features, further comprising an acid/salt low-pH buffering system. 16. The composition of any of the preceding features, wherein the HCA is p-coumaric acid. 17. The composition of any of the preceding features, wherein the composition comprises at least one additional skin care active selected from the group consisting of vitamins, minerals, peptides, sugar amines, sunscreens, oil control agents, flavonoid compounds, antioxidants, protease inhibitors, tyrosinase inhibitors, anti-inflammatory agents, moisturizing agents, exfoliating agents, skin lightening agents, anti-acne agents, anti-wrinkle agents, phytosterols, N-acyl amino acid compounds, antimicrobials, antifungals, and combinations thereof. 18. A skin care essence, comprising: a. niacinamide; b. coumaric acid; c. isopropyl lauroyl sarcosinate; d. pentylene glycol; and e. wherein the composition has a pH of less than about 5.0, a viscosity of about 1 cP to about 15000 cP at 25° C., and is free of HCA crystals. 19. The skin essence of the preceding feature, further comprising sodium sulfite. 20. A method of cosmetically treating skin, comprising: identifying a target portion of skin where treatment is desired; and applying an effective amount of the composition of the preceding feature 1 to the target portion of skin over the course of a treatment period. 21. A low-pH, aqueous skin care composition, comprising: a. a vitamin B3

compound; b. a hydroxycinnamic acid (HCA); c. a polar emollient; d. a co-solvent with a Hansen solubility parameter distance of less than 15 from the HCA; and e. water, wherein the pH of the composition is less than about 5.0 and the composition exhibits less than about 25% HCA degradation according to the HPLC Method. 22. The composition of the preceding feature 21, wherein the composition contains less than 1000 parts-per-million 4-vinylphenol. 23. The composition of the preceding feature 21, wherein the composition exhibits less than about 10% HCA degradation according to the HPLC Method.

(133) The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a functionally equivalent range surrounding that value. For example, a dimension disclosed as “40 mm” is intended to mean “about 40 mm”.

(134) Every document cited herein, including any cross referenced or related patent or application and any patent application or patent to which this application claims priority or benefit thereof, is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

(135) While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover in the appended claims all such changes and modifications that are within the scope of this invention.

Claims

1. A low-pH, aqueous skin care composition, comprising: a. about 0.1% to about 10% of a hydroxycinnamic acid (HCA); b. about 0.1% to about 10% of a polar emollient; c. from about 1.5% to less than 20% of a co-solvent with a Hansen solubility parameter distance of less than 15 from hydroxycinnamic acid (HCA); and d. water, wherein the pH of the composition is about 5.0 or less, wherein the composition meets at least one of the following conditions (i)-(iii): (i) the composition further contains from about 0.1% to about 10% of a vitamin B₃ compound; (ii) the weight ratio between HCA and the polar emollient is from about 1:1 to about 1:10; and (iii) the composition further contains from about 0.1% to 5% of a sensory improving ingredient selected from the group consisting of: sugar alcohols, sugar-based emulsifiers, modified sugar derivatives, and mixtures thereof.
2. The composition of claim 1, wherein the sensory improving ingredient is a sugar alcohol.
3. The composition of claim 1, wherein the weight ratio between sensory improving ingredient and co-solvent is from about 2:1 to about 1:20.
4. The composition of claim 2, wherein the weight ratio between sensory improving ingredient and co-solvent is from about 1.5:1 to about 1:5.
5. The composition of claim 1, wherein the composition is free of HCA crystals.
6. The composition of claim 1, further comprising a low-pH tolerant polymer thickener.
7. The composition of claim 6, wherein the low-pH tolerant thickener is polyacrylate crosspolymer-6.
8. The composition of claim 1, wherein the co-solvent is a glycol selected from the group consisting of propylene glycol, dipropylene glycol, butylene glycol, pentylene glycol, hexylene glycol, ethoxydiglycol, C2-C6 polyethylene glycols, and combinations thereof.

9. The composition of claim 1, further comprising about 0.001% to about 5% by weight of an antioxidant.
 10. The composition of claim 9, wherein the antioxidant is selected from the group consisting of sodium sulfite, sodium bisulfite, sodium metabisulfite, and combinations thereof.
 11. The composition of claim 1, wherein the pH is about 2.0 to about 4.5.
 12. The composition of claim 1, wherein the composition exhibits a Δb^* of less than 6 according to the Color Method.
 13. The composition of claim 12, wherein the Δb^* is less than 2.5.
 14. The composition of claim 1, wherein the HCA is p-coumaric acid.
 15. A skin care essence, comprising: a. niacinamide; b. coumaric acid; c. isopropyl lauroyl sarcosinate; d. pentylene glycol; and wherein the composition comprises a pH of less than 5.0, a viscosity of about 1 cP to about 15,000 cP at 25° C., and is free of HCA crystals.
 16. The skin essence of claim 15, further comprising sodium sulfite.
 17. A method of cosmetically treating skin, comprising: identifying a target portion of skin where treatment is desired; and applying an effective amount of the composition of claim 1 to the target portion of skin over the course of a treatment period.
 18. A low-pH, aqueous skin care composition, comprising: a. a vitamin B.sub.3 compound; b. a hydroxycinnamic acid (HCA); c. a polar emollient; d. a co-solvent with a Hansen solubility parameter distance of less than 15 from the HCA; and e. water; wherein the pH of the composition is less than about 5.0 and the composition exhibits less than about 25% HCA degradation according to the HPLC Method.
 19. The composition of claim 18, wherein the composition contains less than 1000 parts-per-million 4-vinylphenol.
 20. The composition of claim 18, wherein the composition exhibits less than about 10% HCA degradation according to the HPLC Method.
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